

WINDSOR, ON, Canada

WMO: 715380

Lat: 42.2788N

Lon: 82.9494W

Elev: 190

StdP: 99.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.6	-22.0	0.5	-15.9	-18.8	0.7	-12.5	13.9	-3.3	12.3	-3.8	4.9	240	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	32.0	23.2	30.5	22.4	29.1	21.6	24.7	29.7	23.7	28.4	22.8	27.2	5.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.3	27.6	22.2	17.3	26.5	21.4	16.4	25.7	75.9	29.8	71.9	28.6	68.5	27.2	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.1	9.0	DB	-19.4	34.6	4.0	1.7	-22.3	35.8	-24.6	36.8	-26.9	37.7	-29.8	39.0	(4)
(5)			WB	-20.0	26.6	3.8	1.0	-22.7	27.3	-25.0	27.9	-27.1	28.4	-29.9	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.6	-2.4	2.3	8.9	15.4	21.0	23.2	22.2	18.5	11.9	5.2	-0.4	(6)	
	DBStd	10.55	6.01	5.64	5.71	4.84	4.62	3.78	2.99	2.80	3.85	4.57	4.81	5.04	(7)	
	HDD10.0	1612	423	348	247	76	7	0	0	0	1	31	157	323	(8)	
	HDD18.3	3404	681	581	496	285	117	17	1	3	45	205	394	579	(9)	
	CDD10.0	1702	1	1	10	43	173	329	408	378	255	91	13	1	(10)	
	CDD18.3	455	0	0	1	2	26	96	152	123	49	7	0	0	(11)	
	CDH23.3	3418	0	0	4	26	222	763	1227	827	311	39	0	0	(12)	
	CDH26.7	996	0	0	0	3	58	247	393	211	76	6	0	0	(13)	
(14) Wind	WSAvg	4.4	5.2	5.1	5.0	5.1	4.3	3.9	3.5	3.2	3.4	4.2	4.7	4.8	(14)	
(15) Precipitation	PrecAvg	920	54	57	75	82	78	102	88	88	84	58	77	82	(15)	
	PrecMax	1152	111	155	134	162	144	168	250	202	251	110	157	163	(16)	
	PrecMin	649	18	7	27	34	30	28	39	16	18	16	23	20	(17)	
	PrecStd	133	25	35	27	30	29	36	47	49	49	27	37	32	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	15.1	22.3	26.7	30.7	33.4	34.2	32.8	31.5	27.2	19.1	14.6	(19)	
		MCWB	11.3	11.4	16.5	18.4	21.0	23.3	24.7	23.6	22.9	20.1	14.0	13.0	(20)	
	2%	DB	10.1	10.3	17.2	22.7	28.3	31.4	32.1	30.5	29.0	23.9	16.3	11.4	(21)	
		MCWB	9.3	7.9	12.8	15.3	20.0	22.3	23.3	23.2	21.4	18.2	13.1	9.8	(22)	
	5%	DB	6.7	7.2	13.9	19.9	26.0	29.6	30.5	29.0	26.8	21.4	14.5	8.6	(23)	
		MCWB	5.2	5.0	10.2	13.9	19.1	21.5	22.7	22.2	20.2	16.6	11.6	6.8	(24)	
	10%	DB	3.9	4.5	10.7	17.0	23.3	27.6	28.9	27.5	24.9	19.2	12.4	6.1	(25)	
		MCWB	2.7	2.7	7.6	11.8	17.2	20.3	21.7	21.3	19.4	15.1	9.8	4.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.1	16.9	18.9	22.9	24.7	26.4	25.9	24.7	21.5	15.5	13.2	(27)	
		MCDB	12.3	14.1	21.4	25.7	29.1	31.3	31.8	30.7	29.8	25.8	17.2	14.1	(28)	
	2%	WB	9.2	8.4	13.5	16.5	21.2	23.5	24.8	24.5	22.9	19.5	13.7	10.2	(29)	
		MCDB	9.8	10.2	16.2	20.8	26.2	29.5	30.0	28.7	26.5	22.7	16.0	11.3	(30)	
	5%	WB	5.4	5.0	10.8	14.5	19.8	22.4	23.7	23.4	21.6	17.3	11.9	7.0	(31)	
		MCDB	6.5	6.8	13.5	18.9	24.7	27.9	28.5	27.2	25.0	20.5	13.9	8.5	(32)	
	10%	WB	2.8	2.8	7.8	12.4	18.3	21.5	22.8	22.4	20.4	15.7	10.0	4.6	(33)	
		MCDB	3.8	4.3	10.3	16.6	22.5	26.2	27.2	26.1	23.6	18.5	12.2	6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.0	8.3	9.8	10.0	9.5	9.3	8.8	9.2	8.4	6.9	5.7	(35)	
		MCDBR	8.8	10.3	13.0	14.2	12.7	11.2	10.7	10.1	11.5	11.3	9.7	8.3	(36)	
	5% WB	MCWBR	8.2	8.7	9.5	8.9	6.9	5.2	4.8	4.8	5.6	6.4	7.5	7.5	(37)	
		MCWBR	8.7	9.8	12.5	12.8	11.5	10.2	9.6	8.8	9.7	10.1	8.9	8.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.302	0.335	0.353	0.399	0.433	0.448	0.438	0.430	0.389	0.360	0.320	0.312	(40)	
		taud	2.401	2.291	2.343	2.246	2.200	2.202	2.254	2.266	2.356	2.420	2.496	2.419	(41)	
	Ebn at Noon		840	859	889	869	845	831	835	830	841	824	817	792	(42)	
		Edh at Noon	79	103	111	132	142	142	134	128	109	91	72	72	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.33	3.50	4.62	5.42	6.11	6.10	5.30	4.28	2.78	1.73	1.22	(44)	
	RadStd		0.16	0.27	0.34	0.40	0.46	0.38	0.31	0.21	0.37	0.23	0.23	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+88	N/A	

Nomenclature: See separate page